



PARKTOWN BOYS' HIGH SCHOOL
DEPARTMENT OF GEOGRAPHY

GEOGRAPHY

—June PAPER 1—

MEMO

2 June 2025

Examiner: Mr. A Meintjes
Internal Moderator: Mr. D. Grace

TIME: 3 hours

MARKS: 150 Marks

SECTION A – Climate and Weather

QUESTION 1: Short Questions

1.1. Various possible answers are provided for each question. Write down **only the letter** of the correct answer next to the corresponding number.

1.1.1. Cells of low pressure can be defined as ...

- A. A localised elongated line of low pressure such as a cold front.
- B. Areas of differing pressure that extend around the planet going from East to West.
- C. Concentrated areas of low pressure often resulting in thunderstorms.
- D. Areas of differing pressure that extend around the planet going from North to South. (1)

1.1.2. A South-facing slope in South Africa would experience ...

- A. Warmer conditions between December and February.
- B. Colder conditions between December and February.
- C. Consistently warmer conditions throughout the year.
- D. Consistently cooler conditions throughout the year. (1)

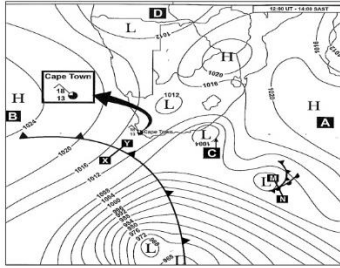
1.1.3. Conditions necessary for frost pockets to form include ...

- A. Clear skies and cold temperatures.
- B. No strong winds at all.
- C. Light rain and no strong winds over the valley.
- D. Only A and B. (1)

1.1.4. Anabatic winds refer to ...

- A. Winds that blow up valley sides and along the valley floor during the day.
- B. Winds that travel vertically up in cities due to excess heat sources.
- C. Sinking winds resulting from temperature inversions.
- D. Winds that blow down valley sides and along the valley floor during the night. (1)

Refer to the diagram below showing a typical synoptic weather map and answer the questions that follow.



1.1.5. The season shown above is most likely ...

- A. Summer.
- B. Autumn.
- C. Winter.
- D. Spring.

(1)

1.1.6. The weather phenomenon labelled C is a ...

- A. Mid-latitude cyclone.
- B. Coastal low pressure.
- C. Tropical cyclone.
- D. Cut-off low.

(1)

1.1.7. The high pressure cell dominating the interior of South Africa has developed because ...

- i. There is a decrease in heat this time of the year.
- ii. The Kalahari High Pressure has increased.
- iii. The interior is cooler than the coastline.
- iv. The cold Benguela current has strengthened.

- A. i & iii ONLY.
- B. None of the above.
- C. All of the above.
- D. i, ii & iii ONLY.

(1)

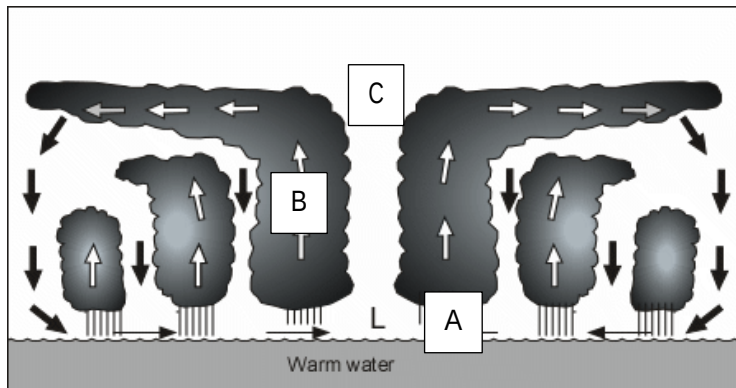
1.1.8. Low pressure systems in **Summer** can lead to ...

- A. Colder conditions over a few weeks.
- B. Clearer skies.
- C. Increased flooding risks.
- D. Increased temperature over a few days.

(1)

(8 x 1) (8)

- 1.2. Refer to the diagram below and answer the questions that follow. For each answer select the correct word in brackets. Write only the question and the correct word. E.g. 1.2.8 Jam.



- 1.2.1. As the cyclone approaches we would experience (warm temperatures and high humidity/low temperatures and low humidity).
- 1.2.2. The eye of the cyclone has the (most powerful winds/caldest winds).
- 1.2.3. The clouds found at B are (alto-nimbus/cumulonimbus) clouds.
- 1.2.4. The air movement at C is (descending/rising).
- 1.2.5. The ocean surface at A will experience a (stormsurge/tidal wave).
- 1.2.6. In order for tropical cyclones to form, they need Coriolis force, hot moist air and intense (high/low) pressure cells forming above.
- 1.2.7. These massive storms typically form towards the (beginning/end) of summer when the ocean is at its warmest.

(7 x 1) (7)

[15]

Question 1.1		Question 1.2	
1	C	1	Warm temp + high humidity
2	B	2	Caldest winds
3	D	3	Cumulonimbus
4	A	4	Descending
5	C	5	Stormsurge
6	B	6	low
7	D	7	end
8	C		

QUESTION 2: Mid-latitude Cyclones

2. Refer to the news article below and answer the questions that follow.

Bomb cyclone? Land hurricane? Powerful storm heading our way evokes interesting names

by FOX 11 Meteorologist Phil DeCastro
Wed, March 12th 2025 at 4:48 PM

Updated Thu, March 13th 2025 at 7:27 AM

At every transitional season, the spring and the fall, powerful mid-latitude cyclones tend to sweep across the United States.

And with the most powerful ones, you start hearing terms like "bomb cyclone" -- or people even likening the storms to hurricanes.

Some of these terms are legit -- some aren't. Let's start with my favorite: the "land hurricane".

First of all, hurricanes form due to a [completely different process](#) than storms like the one tracking across the U.S. late this week. Hurricanes are a [warm-core cyclone](#), driven by a warm ocean and the hot, humid air atop it rising and forming thunderstorms, and the air at their center tends to be warmer than the environment around it.

Storms like the one trekking over land across the U.S. later this week are driven by big horizontal differences in air temperature, and as they strengthen, the air at the center ends up being colder than the environment around it.

They're called [cold-core cyclones](#). So in that regard, this is not going to be a "land hurricane." But I will grant two things.

One, the severe thunderstorms triggered by these mid-latitude cyclones can absolutely produce straight-line winds of hurricane strength, which is anything above 74 miles per hour.

Two, they can sometimes kinda-sorta look alike. It's not perfect, but they both can take on swirling, rotating shapes when viewed by satellite.

Cold-core cyclones, though, do not develop eyes at their center like hurricanes do.



The other term you might hear is "bomb cyclone" and this one is easy enough to tackle. When the central pressure of a cyclone decreases by 24 millibars in 24 hours, it's considered to be "bombing out" or undergoing "bombogenesis". That is, the pressure at the center of the storm is dropping like a bomb. Or, you could say its strength is exploding, like a bomb.

Either way, when you hear a system referred to as a "bomb cyclone" you know it's going to be a powerful one.

- 2.1. Which direction do mid-latitude cyclones travel in the northern hemisphere, west to east or east to west? (1 x 1) (1)

West to east ✓

- 2.2. According to the article, what are the TWO key differences between mid-latitude cyclones and hurricanes? (2 x 2) (4)

Hurricanes are 'warm-core' in that they develop over warm oceans (MUST INCLUDE WARM OCEANS) ✓✓

Mid-latitude cyclones (accept 'cold-core') do not develop eyes ✓✓

- 2.3. What is a 'bomb cyclone' according to the article? (1 x 2) (2)

[Concept]

A really powerful mid-latitude cyclone ✓✓

- 2.4. Differentiate between the Mature Stage and Occulsion Stage of mid-latitude cyclone development. (2 x 2) (4)

[Accept reasonable answers]

Mature stage – Two clear fronts have formed. Clear apex or central low. Strong winds at cold front [accept any ONE] ✓✓

Occlusion stage – Cold front has merged/caught up to warm front. Cold front is undercutting warm front. No apex [accept any ONE] ✓✓

2.5. Draw a simple cross section of a cold front occlusion. Include the following:

A heading for your diagram

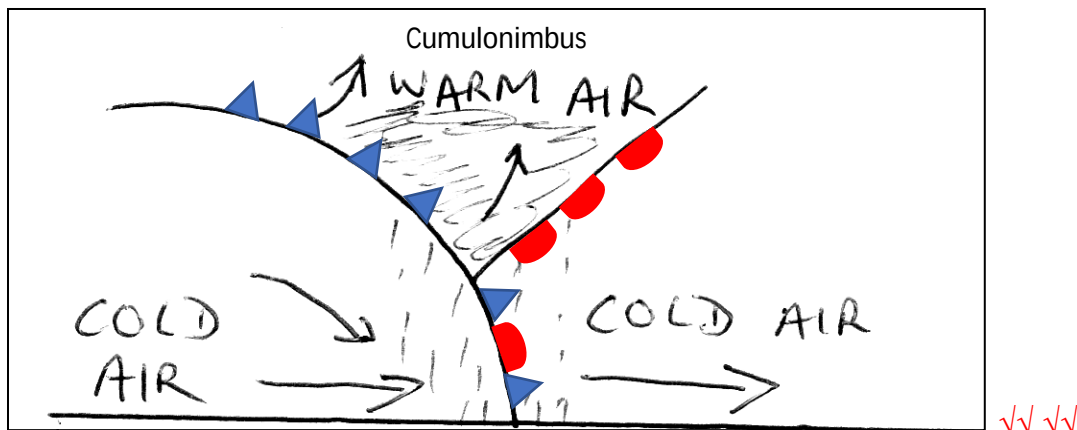
Correct front shapes and symbols

Correct air pocket temperatures

Correct cloud types

(4 x 1) (4)

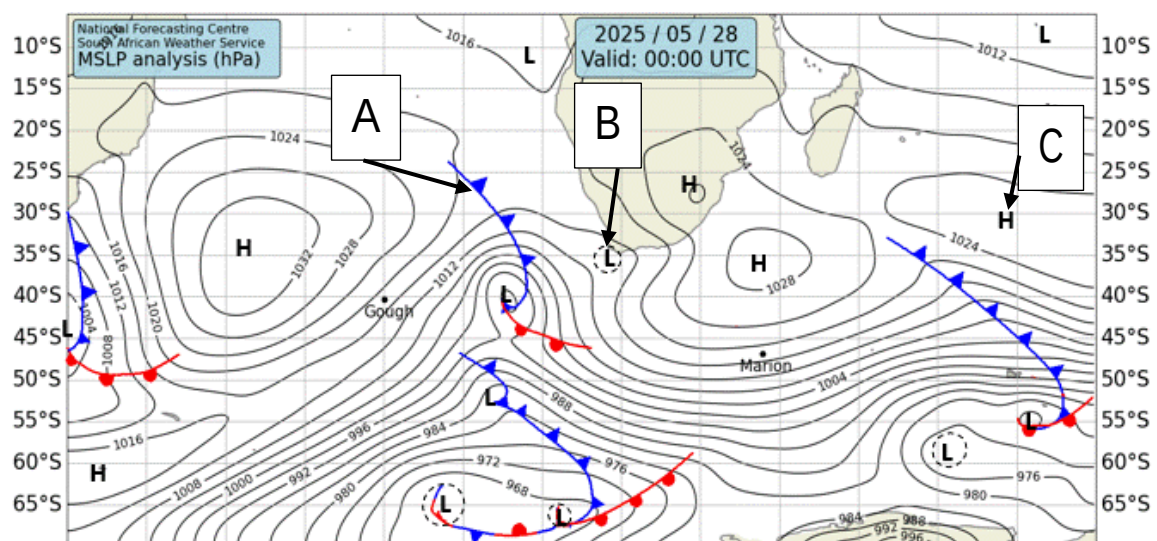
Cold Front Occlusion



[15]

QUESTION 3: Anti-cyclones

3. Study the synoptic weather map below and answer the questions that follow.



3.1. What season is this synoptic weather map depicting?

(1 x 1) (1)

Winter ✓

- 3.2. Provide ONE piece of evidence from the chart to justify your answer to QUESTION 3.1. (1 x 1) (1)

[Accept any ONE]

Date, presence of HP cells over interior, presence of cold fronts ✓

- 3.3. Provide the names for the features labelled A and C on the map respectively. (2 x 1) (2)

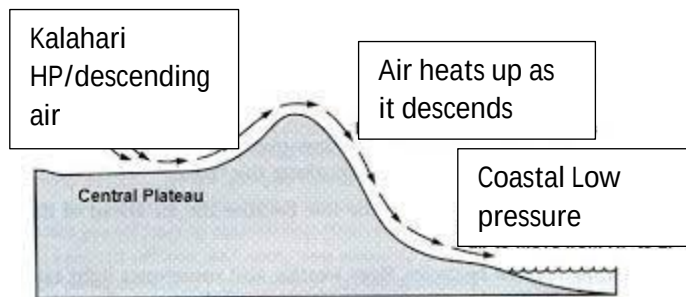
A – Cold front (mid-latitude cyclone)

C – South Indian HP (MUST HAVE “South”) ✓✓

- 3.4. The system labelled B is a coastal low pressure and therefore a travelling disturbance. Which way is it likely to travel, east or west? (1 x 1) (1)

East ✓

- 3.5. Draw a simple cross section showing how the low pressure named at B may lead to the formation of Berg Winds. (4 x 1) (4)



✓✓✓ +1 mark for showing arrows going from HP to LP ✓

- 3.6. Discuss why Berg Winds are a danger to South Africa. Provide TWO reasons in your answer (2 x 2) (4)

[MUST be a discussion: I.e. make point → link to why its dangerous]

Dries out vegetation → Leads to veld fires ✓✓

Dries out crops → threatens livelihoods/food security/food price ✓✓

Can cause heatstroke ✓✓

- 3.7. Describe the role that the Kalahari High Pressure cell plays in the reduction of rain over the interior during this season. (1 x 2) (2)

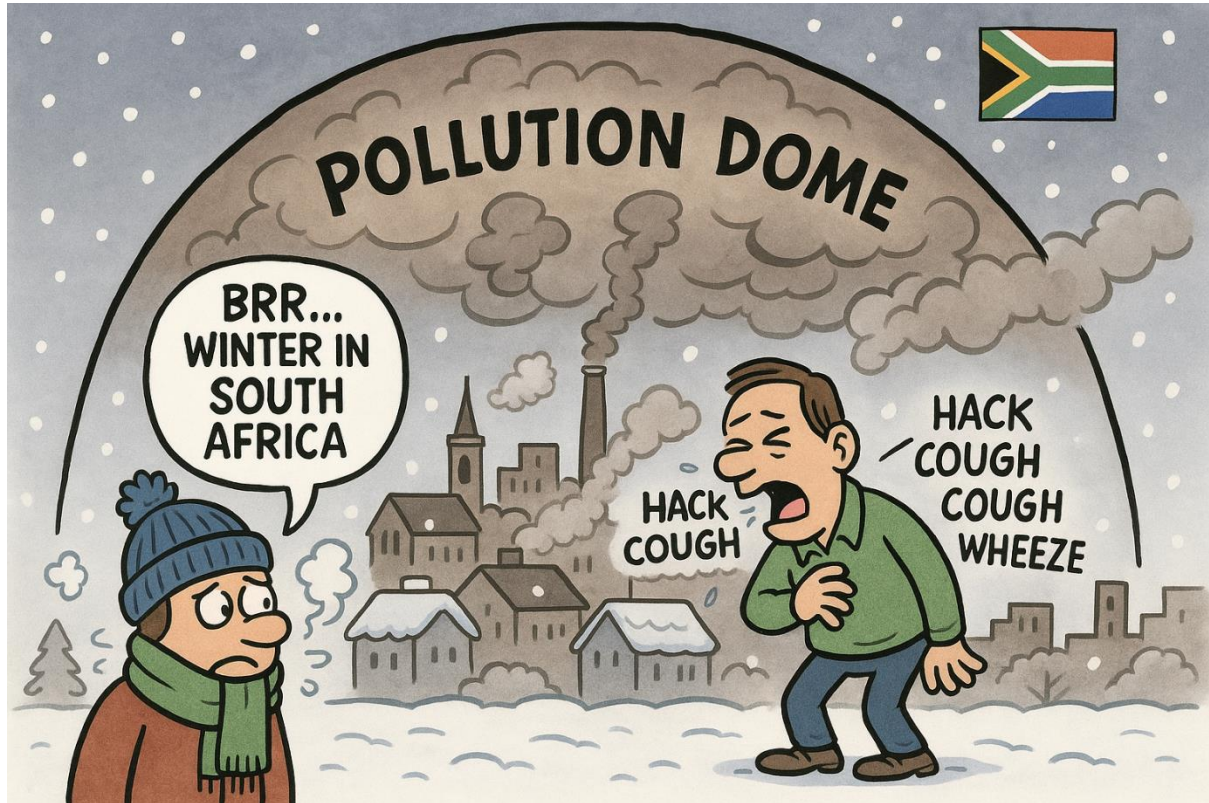
High pressure prevents rising air ✓✓

NOTE: We are not talking about temperature here. Cold resulting in no evaporation = no marks.

[15]

QUESTION 4: Urban Climates

4. Study the ChatGPT image and answer the questions that follow.



4.1. Did ChatGPT get the season that pollution domes form in right? (1 x 1) (1)

Yes ✓

4.2. What is the season depicted in this image? (1 x 1) (1)

Winter ✓

4.3. What characteristic of South African cities did ChatGPT get wrong? (1 x 1) (1)

We don't get thick snow in SA ✓

4.4. Describe TWO factors that lead to the formation of pollution domes. (2 x 2) (4)

[Accept reasonable answers]

✎ Vehicle exhaust fumes ✓✓

✎ Burning of fossil fuels to generate electricity ✓✓

✎ Running of generators and any other machinery that emits noxious gases ✓✓

- 4.5. In a short paragraph, briefly discuss TWO negative economic impacts of pollution domes and TWO solutions that local governments could implement to reduce these impacts. (2 x 4) (8)

[Accept reasonable answers]

Possible Negative Economic Impacts [Any TWO]:

Increased healthcare costs – More people fall ill due to air pollution, leading to higher public and private healthcare spending. ✓✓

Reduced productivity – Sick workers mean fewer working hours and lower economic output. ✓✓

Damage to buildings and infrastructure – Acid rain and soot from pollution domes corrode buildings, increasing maintenance and repair costs. ✓✓

Loss of tourism revenue – Poor air quality and unattractive cityscapes can deter tourists, impacting local businesses and hospitality. ✓✓

Decline in property values – Areas with persistent pollution are less desirable, reducing property demand and tax revenue. ✓✓

Possible Solutions by Local Governments [Any TWO]:

Enforce emissions regulations – Set limits on industrial and vehicular emissions.

Promote cleaner transport – Encourage the use of electric buses, trains, and cycling infrastructure. ✓✓

Introduce congestion taxes – Charge drivers to enter city centres to reduce traffic and emissions. ✓✓

Plant trees and green spaces – Vegetation absorbs pollutants and improves urban air quality. ✓✓

Raise public awareness – Educate citizens on pollution sources and encourage eco-friendly practices. ✓✓

Improve public transport systems – Reliable and affordable public transport reduces the number of private vehicles on the road. ✓✓

[15]

TOTAL SECTION A: [60]

SECTION B - Geomorphology

QUESTION 5: Short Questions

5.1. The following descriptions are all features of a river system. State whether each of these statements is TRUE or FALSE. Write only the question and the answer. E.g. 5.1.8. True

5.1.1. Deposition is dominant at in the lower course of a river.

5.1.2. A transverse river profile shows a general path of a river from the source to the mouth.

5.1.3. Meandering rivers and ox-bow lakes are a features of the upper river course.

5.1.4. Most headward erosion takes place in the middle course of a river.

5.1.5. Oxbow lakes typically form within deltas.

5.1.6. The Nile travelling through the desert would be an example of an episodic river.

5.1.7. Debris in a river moving through traction refers to the process whereby large boulders are removed with machines.

(7 x 1) (7)

5.2. Match the terms in the table with one of the descriptions. Write only the number of the question and the correct term (5.2.1–5.2.8) in the ANSWER BOOK, for example 5.2.9 Sunshine.

Catchment area	Knickpoint	Sinuosity
Valley	Meander	Abrasion
Antecedent drainage	Hydraulic action	Source
Watershed	Confluence	Interfluves
Mouth	Laminar flow	Turbulent flow

5.2.1. This is the point where one river joins another river.

5.2.2. High ground separating one drainage basin from another adjacent drainage basin.

5.2.3. A ridge or area of land dividing two river valleys within a drainage basin.

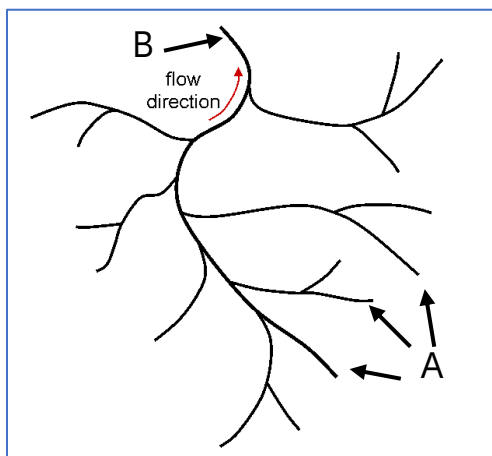
- 5.2.4. A term to describe a particular point in a river or channel where there is a sharp drop in altitude, usually resulting in the formation of waterfalls.
- 5.2.5. A process whereby the continual colliding of water and other materials on the base of the plunge pool gradually deepens the pool.
- 5.2.6. The term that refers to the end of a river.
- 5.2.7. The area from which rainfall flows into a river, lake, or reservoir.
- 5.2.8. A term used to describe calm water with little turbulence.

(8 x 1) (8)
[15]

Question 5.1		Question 5.2	
1	True	1	Confluence
2	False	2	Watershed
3	False	3	Interfluves
4	False	4	Knick point
5	False	5	Hydraulic Action
6	False	6	Mouth
7	False	7	Catchment Area
		8	Laminar flow

QUESTION 6: Drainage basins

6. Refer to the figure below showing a drainage basin and answer the questions that follow.



6.1. Provide labels for river features at A and B respectively? (2 x 1) (2)

A – Tributaries/Source

B – Mouth

6.2. What is the shape of this drainage basin? (1 x 1) (1)

Dendritic

6.3. Determine the stream order of the river exiting at B. (1 x 2) (2)

3rd Order

6.4. Does this shape indicate that this basin *depends* on the geology and topography of the area, or is it *independent* of the geology and topography of the area? (1 x 1) (1)

Independent

6.5. Using the below information, calculate the drainage density of this basin.

Length of streams = 44.5km

Area of basin = 8.23km²

Formula: $\frac{\text{Total length of streams in basin (km)}}{\text{Total area of the drainage basin (km}^2\text{)}}$ (3 x 1) (3)

5.4 km/km²

6.6. There are many factors that can influence drainage density. Briefly discuss any THREE of these factors. (3 x 2) (6)

[Any THREE + DISCUSSION – i.e. Climate, high rain more density]

Evaporation Rates – High rate, less density

Climate – Wet, more dense

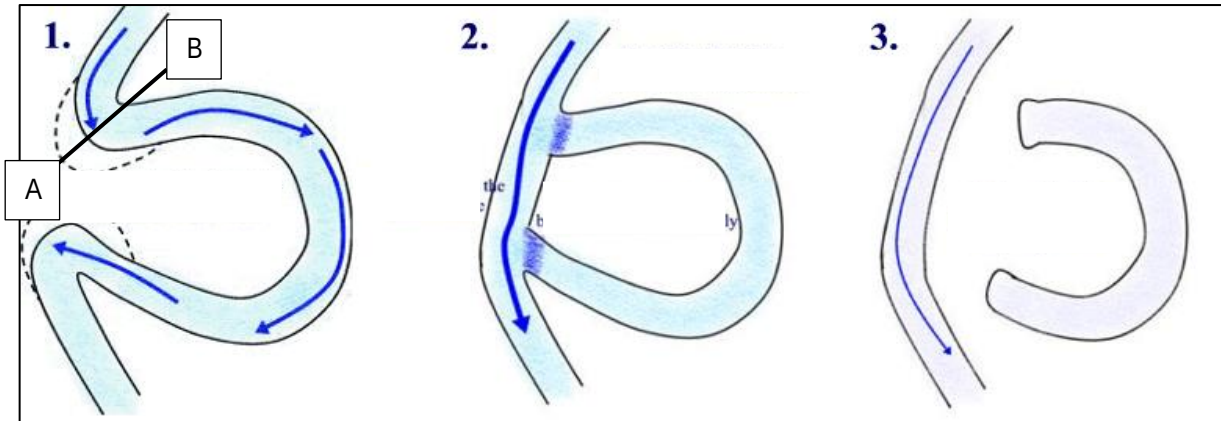
Infiltration – More porous, less rivers/dense

Rock Type – More impermeable, more rivers (more surface water)

[15]

QUESTION 7: Fluvial Processes

7. Study the figure below and answer the questions that follow.



7.1. What feature is being formed in the diagram above? (1 x 1) (1)

Oxbow lake

7.2. In which course of the river are these features most likely to form? (1 x 1) (1)

Lower/middle

7.3. Would the likely gradient for the above diagram be steep or gentle? (1 x 1) (1)

Gentle

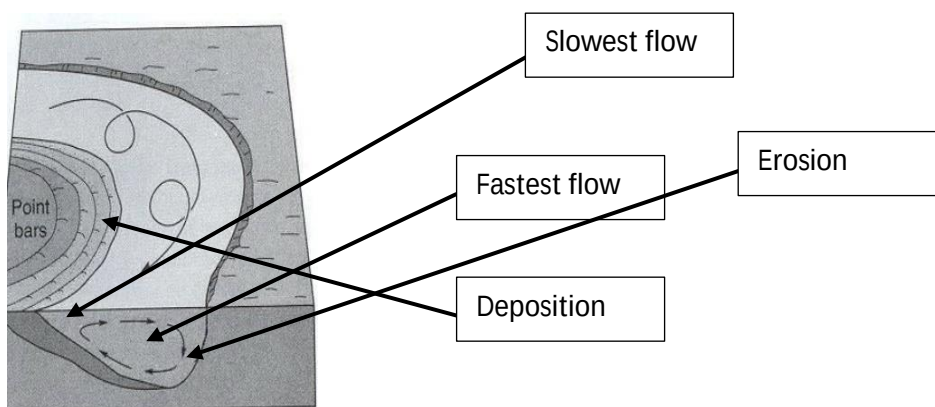
7.4. In diagram 1 the river is meandering. Draw a simple cross section between points A and B showing how the water is flowing in this part of the river. Include the following:

Flow speeds (indicate the strongest flow and slowest flow)

Deposition (point bars)

Erosion (undercutting)

(4 x 1) (4)



7.5. In diagram 1, explain the process occurring at the “narrow neck” of the meander.

(1 x 2) (2)

[Accept reasonable answers]

Increased erosion

Abrasion

Material is being removed and the outer-bank is eroding

7.6. Features like the above often develop as a result of river rejuvenation.

Define the term *River Rejuvenation*.

(1 x 2) (2)

[Concept]

Increase in energy in a river

7.7. Describe TWO negative consequences for people and/or animals that live downstream of a river that has been rejuvenated.

(2 x 2) (4)

[Accept reasonable answers – MUST link to negative]

Flooding – can cause death/crop loss

Increased erosion – topsoil removal

Habitat change – animals/tourism loss

[15]

QUESTION 8: River Management

Study the article below and answer the questions that follow.

One river running through South Africa's richest province is so polluted it is killing people



Once a shimmering centerpiece of the area, Centurion Lake in Gauteng has become, in the words of some local residents, “a pungent ghost town that nobody wants to go near.”

Standing on the banks of what was once a bustling tourist attraction, with vibrant restaurants and shops, the overwhelming smell of raw sewage now dominates the scene.

Lake Centurion is now so filled with sediment and pollution that it no longer functions as a proper lake. Incoming water from the Hennops River is heavily contaminated with sewage, plastic, and toxins.

Water expert Professor Anthony Turton told BusinessTech that Hennops pollution has been a persistent problem, driven by industrial runoff, long-standing sewage discharge, and poor wastewater management.

As a result, adjacent properties like Centurion Mall have suffered falling real estate values and tenant losses.

To date, the river continues to carry its pollution toward Hartbeespoort Dam, amid controversial claims of radionuclide contamination near Pelindaba, which authorities have denied.

Illegal sand mining, especially upstream, has exacerbated sediment buildup, contributing to the lake’s degradation and reflecting a broader national problem.

On top of this, densely populated surrounding areas like Tembisa, Ivory Park, and Olifantsfontein lack essential services such as sanitation, waste removal, and stormwater control.

As a result, the Hennops is continuously burdened with raw sewage, industrial effluent, and litter.

State of the river

A major pressure point is the Olifantsfontein Wastewater Treatment Works (WWTW). Designed to handle 110 million litres per day, it is routinely overwhelmed, sometimes processing nearly double its capacity.

This results in tens of millions of litres of untreated or poorly treated effluent flowing into the river daily.

“The simple truth is that WWTWs are slowly rendering the drinking water of the country unusable,” said Turton.

"This is now a material factor in the dysfunction of the Ekurhuleni Water Care Company plant, along with fats, oil and grease, and of all things, clogging of the machinery," wrote Turton for the Daily Maverick.

Very broadly, these issues have made the Hennops River now one of the most polluted waterways in Gauteng.

A recent study by Thabiso Letseka from the University of the Witwatersrand's Faculty of Science confirmed many fears about the dangers of the river, finding it critically polluted and unfit for drinking, irrigation, or recreation.

The South African Human Rights Commission (SAHRC) found that local mismanagement of wastewater treatment plants has been key in the severe pollution of rivers like the Hennops, threatening ecosystems, health, and livelihoods.

Despite decades of complaints and directives, the SAHRC said that the government has failed to act, violating constitutional rights to a safe environment and proper water services.

"Without sounding dramatic, South Africa is slowly committing ecocide – national suicide by poisoning its own drinking and crop production water," Turton previously told BusinessTech.

8.1. In which province is Centurion Lake located? (1 x 1) (1)

Gauteng

8.2. What is the main source of pollution in the Hennops River? (1 x 1) (1)

Industrial runoff and/or sewerage

8.3. List TWO economic impacts the pollution of Centurion Lake has had on the local economy. (2 x 1) (2)

[Any reasonable list – MUST BE ECONOMIC]

Shops closed

Tourism loss

Property value plummet

Health of labour force compromised – reduced productivity

8.4. How does poor service delivery in areas like Tembisa and Ivory Park affect the health of the Hennops River? (1 x 2) (2)

[Accept reasonable answers]

Pollution in these areas, trickles down into the Hennops River

- 8.5. Which constitutional right is being violated according to the SAHRC? (1 x 1) (1)

Constitutional rights to a safe environment and proper water services

- 8.6. Considering the following quote from the article: "Without sounding dramatic, South Africa is slowly committing ecocide – national suicide by poisoning its own drinking and crop production water." Write a short paragraph in which you explore what the future may hold for South Africans if we do not improve our current situation, and provide solutions that leadership could consider to improve the situation. (4 x 2) (8)

[Accept reasonable answers]

What the Future May Hold for South

1. **Water shortages** – Reduced availability of clean water for drinking, sanitation, and industrial use.
2. **Food insecurity** – Contaminated water affects crop irrigation and livestock, reducing food quality and quantity.
3. **Public health crisis** – Increase in waterborne diseases such as cholera and diarrhoea due to polluted water sources.
4. **Economic decline** – Sectors like agriculture, tourism, and fishing could suffer, leading to job losses and lower GDP.

Solutions to Fix the Water Pollution Problem

1. **Upgrade wastewater treatment plants** – Ensure proper treatment before water is released into rivers.
2. **Enforce environmental laws** – Monitor and penalise industries and municipalities for illegal dumping and pollution.
3. **Public education campaigns** – Promote awareness on how individuals and communities can prevent water pollution.
4. **Protect river catchment areas** – Rehabilitate wetlands and maintain natural vegetation to filter pollutants and reduce runoff.

[15]

TOTAL SECTION B: [60]

SECTION C - Mapwork

RESOURCE MATERIAL

1. 2926AA Bloemfontein (NORTH) topographical map and the 2926 10 Bloemfontein orthophoto

GENERAL INFORMATION ON Bloemfontein



Bloemfontein ([/ˈbluːmfonteɪn/](#) *BLOOM-fon-tayn*^{[4][5]} Afrikaans: [\[ˈblumfontɛɪn\]](#)), also known as **Bloem**, is the capital and the largest city of the [Free State](#) province in [South Africa](#). It is often, and has been traditionally, referred to as the country's "judicial capital", alongside the [legislative](#) capital [Cape Town](#) and [administrative](#) capital [Pretoria](#), although the highest court in South Africa, the [Constitutional Court](#), has been in [Johannesburg](#)^{[6][7][8]} since 1994.

Situated at an elevation of 1,395 m (4,577 ft) above sea level, the city is home to 256,185 (as of 2011)^[9] residents and forms part of the [Mangaung Metropolitan Municipality](#) which has a population of 747,431.^[10] It was one of the host cities for the [2010 FIFA World Cup](#).

QUESTION 9: Calculations

- 9.1. What is located at F on the topographical map? (1 x 1) (1)
- 9.2. Provide the coordinates for communication tower in E1. (2 x 1) (2)
- 9.3. Provide the map code or map reference for the map North of 2926AA. (1 x 1) (1)
- 9.4. Choose the correct option below.

The area covered by the orthophoto map as represented on the topographic map is approximately:

- A) 3145km².
- B) 3.145km².
- C) 3.145m².
- D) 3145m².

(1 x 2) (2)

- 9.5. Calculate the current magnetic declination for 2025. Show ALL calculations. Use the following headings as a guide:

The difference in years:

Mean annual change:

Total change:

Magnetic declination for 2025:

(4 x 1) (4)

[10]

QUESTION 10: Interpretation

- 10.1. Refer to the river network in B1.
- 10.1.1. What type of rivers dominate this area, permanent or exotic? (1 x 1) (1)

- 10.1.2. What are the primary uses of these rivers? (1 x 1) (1)
- 10.1.3. This river system has a low drainage density. Discuss TWO factors that would influence this system's drainage density. (2 x 2) (4)
- 10.2. Refer to the vast amount of industrialisation in Tempe (block E2).
- 10.2.1. How may this industrialisation lead to acid rain in the surrounding areas in Winter? (1 x 2) (2)
- 10.2.2. During Summer the area of Tempe experiences a strong urban heat island effect. Draw a simple diagram showing the formation of an urban heat island. (2 x 1) (2)
- 10.2.3. Suggest ONE way that the people of Tempe can combat the effects of urban heat islands. (1 x 2) (2)

[12]

QUESTION 11: GIS

Refer to the Freestate National Botanical Gardens (B3 & C3) and the extract of information below from website and answer the questions that follow.

The status of biological invasions and their management in South Africa in 2022

	A1: New alien species continue to arrive in South Africa every year through several different pathways. A2: Native and alien species are spread by humans around South Africa. A3: Intentional legal introductions are well regulated; the new National Border Management Authority promises to improve the prevention of illegal and accidental introductions.
	B1: The process of documenting alien species in the country has been substantially improved and is now transparent; this will facilitate management and regulatory decisions. B2: Knowledge of the distribution of alien species has been improved by citizen science and the digitisation of historical records; structured surveillance remains essential to inform management and track trends. B3: Invasive species, in particular trees and freshwater fishes, have 'Major' negative impacts on people and nature across the country.
	C1: Invasions are distributed across the country including in protected areas. C2: The impact of invasions on water resources, rangeland productivity and biodiversity are severe; improved workflows to track these impacts are vital for prioritising interventions.

- 1.1. Provide ONE example of line feature found within the Freestate National Botanical Gardens (Block B3 & C3). (1 x 1) (1)
- 1.2. Is the topographical map an example of raster or vector data? (1 x 1) (1)
- 1.3. Buffering was clearly used before building this park. Provide evidence from the map to justify this statement. (1 x 1) (1)
- 1.4. Consider the information provided above. How would GIS be used to monitor the pathways that invasive species move through? (1 x 2) (2)
- 1.5. When generating reports on these invasive species, would you use data manipulation or buffering when creating these charts? (1 x 1) (1)
- 1.6. Discuss how you would use GIS in the safeguarding of the water supply that runs through the gardens. (1 x 2) (2)

[8]

TOTAL SECTION C: [30]

GRAND TOTAL: 150 Marks